HiLabs Term Mapping Challenge

ICD-10-CM and SNOMED CT mapping engine — Documentation

Team: **VK** | Demo video:
[https://drive.google.com/file/d/1idg8b8HSgZEQU8RDUrm7nQneUD4EPfRo/view?usp=sharing]

1. Approach

The task is to map a CodEx-extracted diagnosis string plus its chunk context to one ICD-10-CM code and one SNOMED CT concept, scored by exact match, with the **lower of the two systems** as the headline. We treat it as **candidate retrieval followed by a learned decision**, built separately for each system. We deliberately do not derive one system from the other, because the two do not align one-to-one; the second system's candidates instead feed the ranker as *coupling features* (how often a candidate co-occurred in training with the other system's best candidates), so disagreement becomes information rather than an error.

Three observations from the data shaped the design. (1) The dictionaries are the legal answer space, and about 98% of gold ICD codes are billable leaves, while the dictionary also holds parent codes whose description is the bare term ("Cough" is R05, but the coder picked R05.9). (2) Gold labels carry local coding convention that no public resource contains, so the training pairs are used as extra synonyms, priors and ranker supervision. (3) Errors concentrate where lexical distance and clinical distance disagree (laterality, acuity, negation, abbreviations), so the final decision is made by a model that sees many signals at once.

2. Architecture

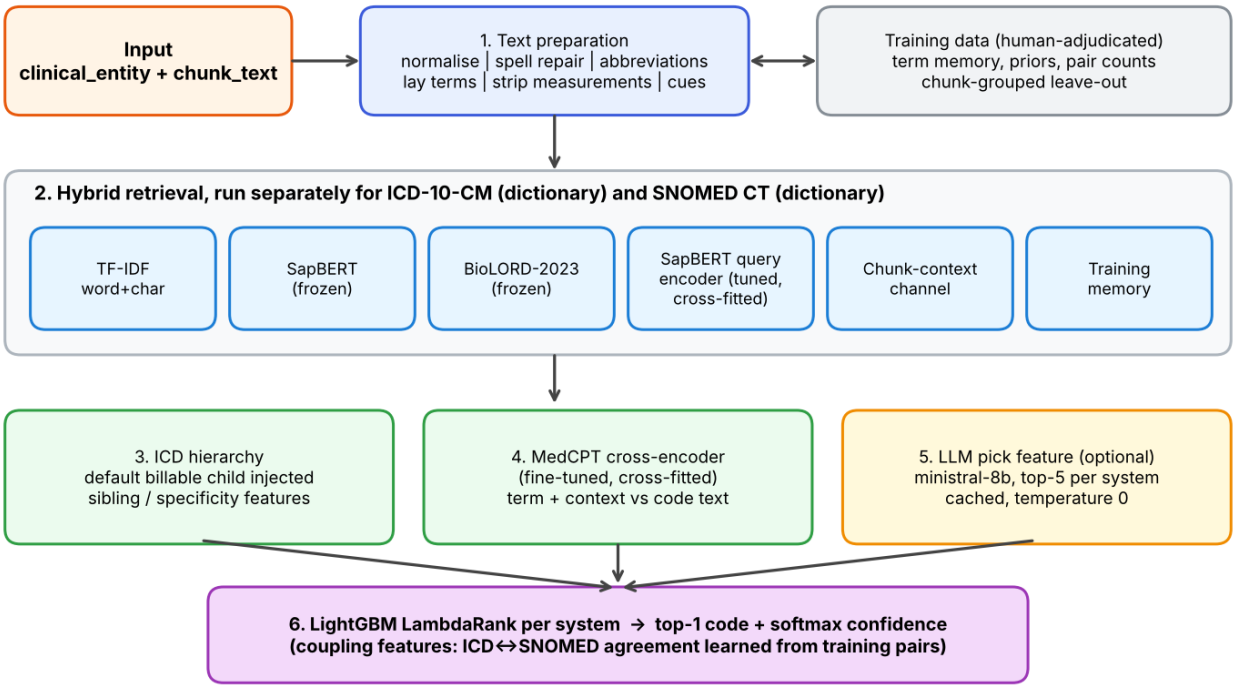


Figure 1: Pipeline. Stages 2–6 run per coding system; every learned component is cross-fitted so the ranker is trained on honest (out-of-fold) scores.

Resource	Where it sits
SapBERT-from-PubMedBERT-fulltext (cambridgeltl)	frozen dense retrieval; also the starting point of the fine-tuned query encoder
BioLORD-2023 (FremyCompany)	second frozen dense channel
MedCPT-Cross-Encoder (ncbi)	fine-tuned on our labels; scores (term + context, candidate text)
ministral-8b-2512 (Mistral API), optional	one pick per term among each system's top-5, used only as a ranker feature
LightGBM, scikit-learn, rapidfuzz, ftfy	LambdaRank reranker, TF-IDF, spelling repair, text fixing
Supplied ICD-10-CM and SNOMED CT dictionaries	legal answer space; UMLS and the official SNOMED-to-ICD map were **not** used

3. Design decisions

- **Hybrid retrieval, union of shortlists.** Lexical scores catch exact and rare terms, dense scores catch paraphrase; the candidate set is the union of every channel's top-K, because the ranker can never recover a code that retrieval missed.
- **Spelling and abbreviation expansions are appended, never substituted.** Replacing words caused regressions (for example *rashes* to *ashes*, *nonintractable* to *intractable*); appended text keeps the original evidence. Corrections are accepted only if they look like a real typo of the same word, and glued words ("chronickidney") are split.
- **Ambiguous abbreviations are left out of the static list** (MS, RA, PE, PID, ...). The chunk is the evidence: definitions written in the chunk ("chronic kidney disease (CKD)") are used, and the ranker and cross-encoder see the context. The billing tag "(HCC)" is stripped rather than read as hepatocellular carcinoma.
- **Specificity policy is explicit, not accidental.** Every parent candidate gets its default billable child injected (R05 → R05.9, I50 → I50.9), and sibling-aware features tell the ranker how many siblings compete, whether an "unspecified" fallback exists and how much more specific a candidate is than the text supports. On training terms that exactly equal a dictionary description, descending to the default child raised the hit rate from 65% to about 90%.
- **Cross-fitting everywhere.** The tuned SapBERT, the cross-encoder, the training memory and the pair/prior counts are computed out-of-fold and chunk-grouped, so no training row (or any row of its chunk) sees its own label. This keeps the ranker's inputs at test-time quality.
- **LLM as a feature, not the decision.** The LLM sees only the top-5 per system plus two similar training examples from other chunks; LightGBM learns how far to trust it. Answers are cached, and the engine runs without it (feature = NaN) if no key is present.
- **Reranker size chosen by chunk-grouped CV** over boosting rounds (100 to 600) rather than fixed; the cross-encoder feature is switched on per system only if it wins on dev.
- **Cue handling.** Negation, history and family-history cues are detected (NegEx-style) and given to the ranker together with the same cues in each candidate's description, plus a cue-aware extra search ("history of ...") when the cue comes from context.

4. Results

Development fold and lockbox (measured)

Exact-match accuracy. The development fold (fold 4) is scored by a model fitted on the other four chunk-grouped CV folds and is used for design decisions; the lockbox (15% of chunks, 362 ICD and 429 SNOMED terms) is scored once by a model fitted on all CV folds. The headline is the **lower** of the two systems.

Split	ICD-10-CM	SNOMED CT	Headline
Development fold (fold 4)	65.3%	64.9%	64.9%
Lockbox (n = 362 / 429)	**72.1%**	**65.0%**	**65.0%**

The lockbox headline is 65.0%, well below the 95% target in the brief; we report it as measured. The ICD-10-CM lockbox figure is 6.8 points above the development fold and the SNOMED CT figures agree to 0.1 points. With a few hundred terms per split each accuracy carries a margin of roughly ±5 points, so we read the two splits as consistent rather than as evidence that one is better. Mean latency on the lockbox run was 749 ms per term (cross-encoder and LLM feature included).

Known codes vs new codes (long tail)

Split	System	Seen codes	Unseen codes	Gap
Lockbox	ICD-10-CM	76.6% (n = 286)	55.3% (n = 76)	21.3 pts
Lockbox	SNOMED CT	77.3% (n = 216)	52.6% (n = 213)	24.7 pts
Dev fold	ICD-10-CM	71.7%	49.1%	22.6 pts
Dev fold	SNOMED CT	76.6%	55.1%	21.5 pts

On the lockbox, about half of the SNOMED CT terms (213 of 429) and about one fifth of the ICD-10-CM terms (76 of 362) have a gold code never seen in training. On those unseen codes the engine still reaches 55.3% (ICD-10-CM) and 52.6% (SNOMED CT), so it generalises beyond memorised codes, but accuracy is 21 to 25 points lower than on seen codes in both systems and on both splits.

Where the right answer gets lost

Split	Measure	ICD-10-CM	SNOMED CT
Lockbox	Gold code in the candidate list (ceiling)	95.9%	94.9%
Lockbox	Ranked 1st (exact match)	72.1%	65.0%
Lockbox	Gold in ranker top-3 / top-5 / top-10	87.3 / 90.3 / 93.9%	80.7 / 83.9 / 89.0%
Lockbox	Never retrieved	4.1%	5.1%
Dev fold	Gold code in the candidate list (ceiling)	92.5%	94.1%
Dev fold	Ranked 1st / 2nd–5th / below 5th	65.3 / 20.1 / 7.2%	64.9 / 19.5 / 9.6%

Closeness of the wrong answers (lockbox)

Measure	ICD-10-CM	SNOMED CT
Exact match	72.1%	65.0%
Same 3-character category / same semantic tag	80.7%	86.0%
Same ICD chapter letter	89.2%	—

Both systems are correct on the same mention for 54.4% of the 338 mentions that have an ICD gold code. SNOMED CT accuracy on those 338 mentions is 69.8%.

5. Error analysis

Main finding. The dominant loss is *ranking*, not retrieval. On the lockbox the gold code is in the candidate list for 95.9% (ICD-10-CM) and 94.9% (SNOMED CT) of terms, but is ranked first for 72.1% and 65.0%. Top-5 reaches 90.3% and 83.9%, so roughly 18 points of each system sit in positions 2–5, and only 4.1% and 5.1% are never retrieved. The development fold shows the same picture (gold in the candidate list for 92.5% and 94.1%, ranked first for about 65%). Recall is therefore not the bottleneck; choosing among near-identical candidates is, which matches the specificity and sibling-code failure modes below. The weaker *generalisation* on unseen codes (55.3% ICD-10-CM, 52.6% SNOMED CT on the lockbox) is the second gap.

The lockbox has 251 wrong predictions: 101 for ICD-10-CM and 150 for SNOMED CT. Breakdown by failure mode (from `outputs/test_errors.csv`):

System	Failure mode	Count
ICD-10-CM	parent / child (specificity)	1
ICD-10-CM	same 3-character category (sibling code)	30
ICD-10-CM	different category	70
SNOMED CT	same semantic tag	90
SNOMED CT	different semantic tag	60

Parent/child errors are almost absent (1 of 101) because the engine injects the default billable child of every parent code; the remaining specificity errors are sibling choices (30% of ICD-10-CM errors: laterality, acuity, type), and 70% are in a different category, where the concept itself was ambiguous or wrong. For SNOMED CT, 60% of errors stay within the right semantic tag, which is typical of choosing between near-synonymous concepts at different depths.

The gold code *was* in the candidate list for 85.1% of the ICD-10-CM errors and 85.3% of the SNOMED CT errors. These are ranking failures and are fixable; the rest are retrieval misses (the gold code never reached the ranker).

Confidence and abstention (development fold)

Accuracy on the most confident share of terms, ranking by the softmax confidence of the top candidate. Keeping only the confident half of the terms lifts accuracy from about 65% to about 88%, so low-confidence terms can be sent to human review or a stronger adjudicator.

Share of terms kept	**100%**	**90%**	**80%**	**70%**	**60%**	**50%**
ICD-10-CM accuracy	65.3%	70.9%	75.2%	80.5%	84.1%	88.7%
SNOMED CT accuracy	64.9%	69.5%	73.6%	78.7%	82.4%	87.3%

Examples

System	**Entity**	**Predicted**	**Gold**
[–]	**[from test_errors.csv]**	[–]	[–]
[–]	[–]	[–]	[–]
[–]	[–]	[–]	[–]

Failure modes and how the engine treats them

- **Specificity (parent / child / sibling).** The hardest ICD decision; coders' habits inside a category are learned from training counts, but the text often cannot settle laterality, acuity or type.
- **Abbreviation collision with thin context.** Handled through chunk definitions and the cross-encoder, but a one-line chunk gives little to disambiguate.
- **SNOMED granularity and synonym density.** Several valid concepts at different depths; the training convention is the main signal, so unseen concepts are weakest.
- **Long tail.** Codes absent from training depend on dictionary retrieval and the cross-encoder alone.
- **Label noise.** The same normalised term text sometimes has several gold codes in the labelled data; that sets a ceiling for any model (the notebook appendix computes the majority-label ceiling).

What we would do next: a listwise LLM adjudicator over the top-5 candidates only on low-confidence terms (about 18% of terms have the gold code in positions 2–5, so this is the highest-leverage step, bounded by the 95.9%/94.9% candidate ceiling on the lockbox), a self-hosted open-weight model for that step so no text leaves the environment, SNOMED-to-ICD map features once a UMLS licence is available, and a dedicated specificity head trained per ICD category.

6. Innovations

- Cross-fitted learned retriever and cross-encoder feeding a LightGBM ranker, with chunk-grouped leave-out so train-time features match test-time quality.
- Explicit ICD specificity machinery: default-child injection, sibling-aware features and per-category coder-habit priors.
- Coupling features: ICD and SNOMED are predicted independently, and their agreement (learned from training pairs) is a ranker input and a confidence signal.
- Cue-aware retrieval for negated, historical and family-history mentions; lay-language and OCR-glue repair, all appended so original evidence is preserved.
- Confidence (softmax over candidates) per prediction, so low-confidence terms can be routed to human review.

7. Challenges and learnings

- **Replacing words was worse than appending them** (spelling repair and abbreviation expansion); both now append.
- **Train/test mismatch.** A training row could look up its own chunk-mates in the memory, priors and pair counts, which test rows cannot; leaving out the whole chunk (not just the row) makes train-time features match test-time conditions.
- **Cross-encoder base.** A randomly initialised head on SapBERT was replaced by the pretrained MedCPT cross-encoder; the cross-encoder is enabled per system only when it wins on dev.
- **Chunk context in the tuned query encoder** was tried and left off: the term alone was verified better, and context is used in separate channels instead.
- **Fixed 300 boosting rounds** were replaced by CV-chosen rounds.
- **Data hygiene.** Scientific-notation SNOMED ids from Excel are repaired against the dictionary, and gold codes outside the dictionary are excluded from the memory because they can never be scored correct.
- **Cost and privacy.** The only per-term external call is one small-model request (about 240 characters of context). It is optional, cached and can be replaced by a self-hosted model. Mean latency was 749 ms per term on the lockbox run, dominated by the cross-encoder and the LLM request.

8. Reproducibility

Single command:

```
python run.py predict -input data/evaluation_input.csv
       -output outputs/predictions.csv
```

Seed 42; LightGBM deterministic; LLM answers cached. Saved weights are pulled from the Hugging Face Hub (krhk/Hilabs-term-mapping-weights) automatically. See `README.md`.